Transportation's Systems 20-5751

Project Part 5: User Equilibrium Traffic Assignment

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1 Introduction

In this project, students will implement a complete **User Equilibrium (UE)** traffic assignment model using the **Convex Combination (Frank–Wolfe)** algorithm on the Sioux Falls transportation network.

This project combines concepts from:

- Transportation network modeling
- Shortest path algorithms
- Traffic assignment
- Convex optimization
- Numerical optimization
- Exact line search methods

The goal of this project is to develop a fully functional traffic assignment engine capable of computing equilibrium flows on a congested transportation network.

Students are expected to:

- Build the optimization framework
- Implement the iterative UE algorithm
- Perform exact line search using Golden Section Search
- Analyze convergence behavior
- Interpret transportation engineering results

2 Background

2.1 Wardrop's First Principle

Wardrop's First Principle states:

> "At equilibrium, the travel times on all used routes between an origin and destination are equal and less than or equal to those that would be experienced on any unused route."

This condition is known as the **User Equilibrium (UE)** condition.

At equilibrium:

- No traveler can reduce their travel cost by unilaterally changing routes.
- Traffic distributes itself across the network according to congestion levels.

2.2 Beckmann Formulation

The User Equilibrium problem can be formulated as the following convex optimization problem:

$$\min Z(x) = \sum_a \int_0^{x_a} t_a(w)\, dw$$

where:

- x_a = flow on link a
- $t_a(x_a)$ = travel time function on link a

2.3 BPR Link Performance Function

In this project, link travel times must be modeled using the BPR function:

$$t_a(x_a) = t_a^0 \left(1 + 0.15 \left(\frac{x_a}{c_a}\right)^4\right)$$

where:

- t_a^0 = free-flow travel time
- x_a = current link flow
- c_a = link capacity

3 Project Objectives

The objectives of this project are:

1. Implement the Frank–Wolfe (Convex Combination) algorithm
2. Solve the User Equilibrium assignment problem
3. Perform exact line search using Golden Section Search
4. Analyze convergence behavior

4 Network and Data

Students must use the previously implemented Sioux Falls network.

The network should contain:

- Nodes with positions
- Directed links
- Link lengths

The implementation should use the `NetworkX` library.

5 Synthetic Data Generation

5.1 Random Seed

Each student must generate a unique random seed using their student ID:

$$seed = int(\text{hashlib}.sha256(\text{student\_id}.encode()).hexdigest(), 16)\% int(1e8)$$

All generated network parameters must be reproducible using this seed. In the following sections, all random variables must be generated using this seed.

5.2 Free-Flow Speed Generation

For each link ab, generate a free-flow speed:

$$v_{ab} \sim U(40, 100)$$

where:

- v_{ab} is in km/h
- U denotes a uniform distribution

Then compute free-flow travel time:

$$t^0_{ab} = \frac{L_{ab}}{v_{ab}}$$

where:

- L_{ab} = link length
- t^0_{ab} = free-flow travel time

5.3 Capacity Generation

For each link, generate capacity:

$$c_{ab} \sim int(U(1000, 4000))$$

where:

- c_{ab} is measured in vehicles per hour

5.4 OD Demand Matrix Generation

For every OD pair (o, d):

- With probability 0.3, demand exists
- Otherwise demand is zero

If demand exists:

$$q_{od} \sim int(U(100, 1000))$$

where:

- q_{od} is measured in vehicles per hour

5.5 How To Get Them

You must at the first initialize the random seed for numpy.random. Then, you can generate the free-flow speeds, capacities, and OD demands using the specified distributions. Make sure to assign the generated parameters to the correct links and OD pairs in your network model and do not change the order of assigning parameters to links. This will ensure that your results are reproducible and consistent with the project requirements.

6 Required Algorithm

6.1 Overview

The User Equilibrium assignment must be solved using the **Frank–Wolfe / Convex Combination** algorithm.

The implementation must include:

1. Initial feasible solution
2. Cost update

3. Auxiliary flow generation
4. Exact line search
5. Flow update
6. Convergence check

6.2 Step 0: Initial Feasible Solution

1. Initialize all link travel times using free-flow travel times.
2. Perform an All-Or-Nothing assignment.
3. Store the resulting flows as:
$$x^0$$

6.3 Step 1: Update Link Costs

At each iteration, update all link travel times using the BPR function:

$$t_a(x_a) = t_a^0 \left(1 + 0.15 \left(\frac{x_a}{c_a}\right)^4\right)$$

6.4 Step 2: Direction Finding

Using the updated link costs:

1. Perform an All-Or-Nothing assignment
2. Obtain the auxiliary flow vector:
$$y^k$$

6.5 Step 3: Exact Line Search

The step size λ must be obtained using **Golden Section Search**.

Students must minimize:

$$\phi(\lambda) = Z(x^k + \lambda(y^k - x^k))$$

subject to:

$$0 \leq \lambda \leq 1$$

The implementation of Golden Section Search from previous assignments must be reused.

6.6 Step 4: Flow Update

Update link flows using:

$$x^{k+1} = x^k + \lambda_k (y^k - x^k)$$

6.7 Step 5: Convergence Check

The algorithm must stop when one of the following criteria is satisfied:

- Maximum number of iterations reached
- Flow change norm below threshold

One possible convergence measure is:

$$\|x^{k+1} - x^k\|$$

7 Implementation Requirements

The project must be implemented in Python.
The use of the following libraries is allowed:

- NetworkX
- NumPy
- Pandas
- Matplotlib

Students are **NOT** allowed to use:

- Ready-made traffic assignment libraries
- Built-in Frank–Wolfe implementations

8 Required Analysis

8.1 Convergence Analysis

Students must generate the following plots:

1. Objective function vs iteration

2. Step size vs iteration
3. Total system travel time vs iteration

Each plot must include:

- Proper axis labels
- Titles
- Legends when necessary

8.2 Traffic Analysis

Students must analyze:

- Most congested links
- Volume-to-capacity ratios

8.3 Comparison with Previous Assignment Methods

Students must compare:

- All-Or-Nothing Assignment
- Incremental Assignment
- User Equilibrium Assignment

The discussion should include:

- Congestion distribution
- Network performance

9 Sensitivity Analysis

Perform at least TWO of the following experiments.

9.1 Scenario A: Increased Demand

Increase all OD demands by:

- 10%
- 25%
- 50%

Analyze:

- Congestion growth
- Convergence behavior
- Changes in critical links

9.2 Scenario B: Capacity Reduction

Reduce the capacity of all links by 0.1, 0.25, 0.5.
Analyze:

- Congestion
- Convergence behavior
- Changes in critical links

10 Deliverables

Students must submit:

1. Source code
2. Technical report (PDF)
3. Figures and plots

11 Report Structure

The report should contain:

1. Introduction
2. Mathematical formulation
3. Implementation details
4. Results
5. Convergence analysis
6. Transportation interpretation
7. Conclusion

12 Important Notes

- Figures must be readable and professionally formatted.
- Numerical stability and convergence behavior should be discussed.
- Students are encouraged to validate intermediate results during development.